STRAT-TRACE

Source-tagged ^{7}Be for process-aware aerosol model evaluation

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1. Excellence

1.1 Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)

Scientific contribution. STRAT-TRACE will establish when joint observations of airborne and deposited ^{7}Be can distinguish changes in upper-atmospheric supply from changes in wet removal. Controlled DEHM experiments will link each explanation to a verified source budget and test its support across observed years and sampling regimes.

Air-quality model development needs tests that explain why a simulation agrees with measurements. Using the existing beryllium-7 (^{7}Be) implementation in the Danish Eulerian Hemispheric Model (DEHM), STRAT-TRACE will deliver a reusable benchmark showing which process explanations the observations support, exclude or leave indistinguishable. The objectives progress from verified source budgets (O1) to discrimination of supply–removal explanations (O2) and transfer across years and observing regimes (O3).

For particle-bound ^{7}Be, production, boundary supply, transport, radioactive decay and deposition jointly determine the measured signal [1–3]. Air concentration and deposition provide complementary constraints, while production uncertainty and precipitation regime shape their interpretation [1–4]. Source labels quantify model contributions; independent observations test whether their responses can be distinguished after accounting for sampling and uncertainty.

State of the art and the specific advance

Established capability	What STRAT-TRACE will add
Joint ^{7}Be concentration–deposition evaluation [1,2] and improved production formulations [3].	A controlled test of when joint observations reject compensating supply–removal explanations, including nuisance-source uncertainty.
Source tagging already exists; GEOS-Chem includes a stratospheric-source Be7s tracer [16].	A verified multi-origin budget linked to discriminatory power at the actual measurement intervals and its dependence on observing regime.
Controlled removal contrasts [5] and transport-model comparisons [6].	A compact within-DEHM experiment set with identical meteorology and independent evaluation, yielding a protocol reusable in other modelling systems.

Table 1. Novelty is the integrated, observation-aware test of process explanations; individual tracer and statistical tools are established.

The benchmark will quantify how reliably observations distinguish halved or doubled wet-removal rate coefficients from compensating changes in tracer supply, targeting ≥80% recovery at ≤5% false discrimination in synthetic tests. It will identify useful measurement combinations and regimes. DEHM's contribution to the European CAMS regional air-quality system [17] provides a direct route for developers to use these diagnostics.

Measurable objectives, hypotheses and success criteria

Objective / delivery	Testable hypothesis and evidence
O1 — Source-resolved response By M9: verified source contributions and budgets.	H1: wet-removal changes produce origin-dependent responses. Compare source fractions and absolute losses across years and solar phases under matched variable/fixed production. Test whether contrasts exceed numerical and sampling uncertainty.
O2 — Discrimination of explanations By M15: supply–removal response library.	H2: joint air–deposition observations discriminate scenarios that air-only evaluation leaves unresolved. Compare recovery at the same controlled false-discrimination rate and quantify the added information from deposition.
O3 — Transfer and observing value By M21: held-out tests and discrimination map.	H3: budget-supported explanations retain joint skill across held-out years and regimes. Fixed-production controls test source dependence. Assess transfer using separate air/deposition scores and pre-specified uncertainty tolerances.

The hypotheses compare wet removal with alternative explanations involving production, upper and lateral boundary supply, and measurement representativeness. Success requires verified response budgets and a reproducible assessment of observational support. Where explanations remain indistinguishable, the benchmark will quantify the tested discrimination limits and the additional observing information needed to separate them.

Existing implementation and observational preparation

Chham and Ye have implemented ^{7}Be in DEHM. Their EGU 2026 abstract documents production mapping, WRF/ERA5 meteorology, aerosol-bound treatment and surface evaluation [7]. The applicant's preparatory comparison of model resolutions motivates the controlled experiments:

DEHM: 25 km relative to 75 km	Median response
Surface ^{7}Be concentration	+5.9%
Bulk deposition: wet + dry	+13.6%
Precipitation	−45.9%

Table 2. Applicant's preliminary JJA analysis, 2009–2024, 79 model sampling locations. Each response is 100 × (fine/coarse − 1) from paired seasonal means, followed by a spatial median. Locations may share grid cells. These model contrasts define tracer supply and removal exposure as processes to test.

Greater deposition with lower precipitation identifies tracer availability and removal exposure as processes to test. STRAT-TRACE will test these explanations through controlled interventions and independent observational evaluation, extending the existing implementation and evaluation work to process discrimination.

The European air archive was obtained primarily through the European Commission's Joint Research Centre (JRC), retaining national-programme attribution [22]. It includes over 54,000 records from 83 named series with checked values, intervals and units in 2009–2024. The joint archive covers 43 European stations with airborne and rainwater ^{7}Be measurements. Deutscher Wetterdienst (DWD) supplies aerosol, deposition and station/method metadata; 2009–2024 remains the recent joint baseline [20].

Four air series occur in 1986: Berlin, Braunschweig, Risø and Wien-Hohe Warte, with broader coverage from 2006. Chham and Ye have signed a Comprehensive Nuclear-Test-Ban Treaty Organization (CTBTO) agreement to extend air coverage internationally; delivery is pending [23]. WP1 verifies paired windows, station continuity and overlaps; DWD historical recovery continues.

A focused design for a 24 month fellowship

The project uses DEHM alone, with reduced chemistry and AU-confirmed computational resources supporting matched experiments over 1986–2024/2025. Variable and fixed production distinguish source variability from atmospheric response across three complete solar cycles. Long station records support temporal tests, while the wider network supports spatial and regime comparisons. The bounded design uses the existing implementation and measurements, with three wet-removal settings and common forcing. It tests process robustness across historical conditions; climate-change attribution is outside its scope.

1.2 Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices)

One physical tracer with explicit source and boundary accounting

DEHM uses the team's reduced-chemistry WRF/ERA5 workflow at 75 km, with 29 layers and a top near 100 hPa [7]. Transport, mixing, decay and wet/dry deposition are retained. Code, operators, aerosol treatment and CRAC:Be formulation [3,9] are documented and fixed across experiments. Observation-facing runs use time-varying production driven by archived solar-modulation and geomagnetic inputs. Layer 1 is the surface.

Production P(x,t) is partitioned by non-negative masks w_k, with $\Sigma_k w_k = 1$ and $P_k = w_k P$. Labels share transport, decay and deposition, recording production region or boundary entry rather than subsequent altitude or last tropopause crossing.

Label	Source partition and interpretation
UPPER / MIDDLE / LOWER	Production at $p < 300$ hPa; $300 \le p < 700$ hPa; and $p \ge 700$ hPa, respectively, within atmospheric cells. These are diagnostic bands, not three separate meteorological systems.
TOP / LATERAL	Realised upper and lateral exchanges, separately budgeted from production within resolved cells. TOP follows the prescribed upper concentration; boundary additions and withdrawals are accounted for.
INITIAL	Initial inventory propagated through spin-up. Its residual influence is tracked rather than silently attributed to subsequent production.

Table 3. Six labels plus an independent total for each production treatment. A short boundary-shift test checks the pre-specified pressure cut-offs.

At the upper boundary, $n_{top} = Q_{CRAC}(100\ hPa, \varphi, t)/\lambda_7$ is prescribed in atoms kg^{-1} of air; Q is local production in atoms kg^{-1} s^{-1} and λ_7 is the decay constant in s^{-1}. This local production–decay equilibrium approximation retains latitude and solar dependence. TOP tracks the realised boundary contribution separately from in-domain production. Production and circulation above the ceiling remain unresolved; UPPER + TOP is not the complete stratospheric contribution, since pressure bands and entry routes do not establish origin.

Fixed-production companions use a time-invariant CRAC:Be field retaining latitude–altitude structure; TOP uses this field divided by λ_7. LATERAL repeats an annual concentration climatology; meteorology-dependent exchanges are budgeted. Uncontrolled lateral variability restricts attribution to in-domain labels and TOP.

Within each wet setting, variable- and fixed-production partners share meteorology, transport and removal operators. Matched-calendar interannual contrasts test transport and removal together; wet-setting contrasts test removal. Lateral seasonality remains. Normalized diagnostics are checked against fixed-source results for spatial and lagged source effects [3].

Numerical verification and process budgets

Labels must reconstruct total concentration, deposition and burden, including after reweighting. Verification covers units, limiters, boundary updates and nested-label transfer, separating boundary inventory changes from resolved production. The proposed tolerance is below 0.1% above a declared absolute floor and ten times smaller than the smallest interpreted response.

For each label k and common control volume, the inventory N_k (total atoms) will satisfy:

$$dN_k/dt = P_k + F_{in,k} - F_{out,k} - \lambda N_k - Lwet_{,k} - Ldry_{,k} + \varepsilon_k.$$

P and F are production and transfer rates, L denotes removal and ε the residual. Time-step ledgers retain origin labels across pressure bands and pair internal transfers with equal and opposite entries, including moving-boundary/rebinning terms. Boundary inventory adjustments and operator losses are recorded without double counting resolved production. Whole-domain closure is mandatory; unresolved local terms are reported explicitly.

After budget verification within the declared tolerance, simulated concentrations and deposition are matched to actual sampling intervals. Tests using observed coverage and uncertainty then assess whether prescribed supply and wet-removal changes can be distinguished.

Outputs are absolute C and D, source fractions C_k/Ctotal and D_k/Dtotal with denominator floors, and integrated wet losses by origin and pressure band. The removal-rate diagnostic ∫Lwet dt / ∫N dt has inverse-time units.

A fixed experiment matrix over three complete solar cycles

1986–2024/2025 is the principal simulation window: 39 or 40 calendar years, ending in 2025 where consistent inputs permit, otherwise 2024. It spans complete solar cycles 22–24 (September 1986–December 2019) and the rising and maximum phases of the ongoing cycle 25 [21]. All settings share meteorology and years. Records from 1986–2005 support temporal tests across solar phases within verified station segments; the denser later network supports spatial transfer and joint concentration–deposition tests centred on the 2009–2024 deposition baseline. WP1 verifies input continuity and observational coverage; the contingency in Section 3.1 preserves matched experiments.

Case	Intervention	Coverage / interpretation
R + tags	Time-varying CRAC:Be production and linked upper-boundary concentration; reference process operators.	1986–2024/2025. Reference budgets and direct comparison with available observations.
W− / W+ + tags	Multiply the reference aerosol wet-removal rate coefficients by αw = 0.5 / 2; reference αw = 1.	Same full period and production forcing as R. Prescribed stress contrasts test sensitivity; they are not calibrated improvements or confidence limits.
S contrasts	Reweight UPPER and TOP after linearity tests, preserving production–TOP links for source scenarios; TOP-only contrasts test the boundary approximation.	Matched-tag reconstruction over the full period, separately for each source treatment. Supply changes; transport operators do not.
Fixed-source partners	Repeat a common production field and its linked TOP concentration; control LATERAL variability as specified above.	All three wet settings, 1986–2024/2025. Interannual atmospheric response under common source forcing.
Verification controls	Source linearity, concentration/production units, boundary updates, pressure-band and spin-up checks.	Bounded pilots. Spin-up precedes the analysed 1986 start; INITIAL influence is checked.

Table 4. Three wet settings over 39/40 years, each with variable- and fixed-production tracers. Their execution and species costs are accounted for in Section 3.1; source reweighting requires verified linearity.

Wet interventions scale rate coefficients before integration; actual removed mass enters the deposition ledger with positivity maintained. Within each production treatment, production, prescribed boundary concentrations, source masks, decay, dry deposition, meteorology and transport operators are identical across wet settings. The contrasts test represented removal strength rather than an individual scavenging sub-process. WP1 verifies the implementation point and archives the operator tests.

Supply tests treat production and lateral-input uncertainty as alternative explanations. Source reweighting uses each wet setting's verified label fields, retaining its transport and removal response. A CRAC:Be production change at 100 hPa also changes TOP through the equilibrium relation; TOP-only changes test that boundary approximation. A difference of differences in the same output measures the supply–removal interaction, showing how altered supply changes exposure to the sink.

Interannual robustness and independent temporal evaluation

Before fitting, seasons will be classified by rainfall amount, wet-day frequency, persistence and mixing diagnostics. Matched variable- and fixed-source experiments test wet/dry years and solar phases across 1986–2024/2025. Long station records support within-site temporal comparisons; the broader network supports spatial and regime transfer. Stable-station and stable-method subsets, common observed intervals and fixed aggregation weights will distinguish atmospheric changes from changes in network composition. Results retain the geographic and temporal scope of the observations supporting them.

Whole-year and multi-year blocks are assigned to selection and locked evaluation using meteorology, solar phase, coverage and collector-method eras before fitting. All wet settings and source treatments cover both sets; realistic-source runs are scored against observations. Earlier and recent records enter on their valid windows, with joint tests restricted to paired coverage. Later-received records follow a pre-registered extension without retuning. Spatial groups account for shared cells, and whole-year resampling tests regime dependence. Daily rainfall supports daily/weekly, rather than sub-hourly, diagnostics.

Measurements, discrimination tests and interpretation

CTBTO is expected to provide daily airborne ^{7}Be records from around 80 sites, mainly since 2000, including Europe, North America and Asia. WP1 verifies delivered station-years within DEHM coverage and checks JRC/DWD overlaps using station histories and actual sampling windows. Daily records support event-scale concentration tests. DWD's 2018 collector transition remains a method stratum [20], with stable-method subsets checked separately. Joint tests use valid air–deposition pairs from the 43-station European archive; additional air-only sites test spatial transfer. This inventory is separate from the 79 preliminary model locations.

Modelled atoms kg^{-1} are converted to air activity using the decay constant and matching air density. The observation operator averages activity over collection windows, flow-weighted when possible, in mBq m^{-3}, and integrates deposition over its own windows in Bq m^{-2}. Fluxes match the documented collector response, including wet-plus-dry deposition for bulk collectors [20]. Older Bq L^{-1} records require matching rain depth (L m^{-2}); method, uncertainty and decay conventions are retained. Collector–grid differences enter representativeness error. Nondetections are censored; missing values are not zeros.

Realistic, time-varying CRAC:Be responses will be evaluated using air only, deposition only and both. Fixed-production companions diagnose source dependence. Known scenarios generate synthetic measurements with actual gaps, sampling intervals and uncertainty, including cases outside fitted source assumptions. Tests incorporate measurement and representativeness errors, temporal/spatial dependence and model discrepancy; rainfall-related covariance is retained when converting deposition units.

Sensitivity vectors scaled by uncertainty and their singular values first identify responses that are hard to distinguish. Low-dimensional likelihood profiles [8] then test source amplitudes and nuisance terms separately at each wet setting. The discrimination map reports recovery probabilities for prescribed contrasts and detectable source-amplitude bounds. Synthetic tests target ≥80% recovery at ≤5% false discrimination, with simulation uncertainty. Where this is not achieved, results report the measured recovery and the contrasts that remain indistinguishable.

Parameters, regime definitions and scoring rules will be fixed before unblinding held-out tracer responses. Separate air/deposition errors, joint predictive scores and interval coverage will be compared with R using whole-year blocks and method-stratified checks. A candidate is supported only if its intervention has a verified budget effect and its improvement transfers across held-out years without a compensating deterioration beyond the pre-specified uncertainty tolerance. Otherwise it is rejected or classified as indistinguishable among the tested alternatives.

Interdisciplinarity, open science and research-content diversity

Environmental radioactivity determines production, decay and collector conventions; transport physics supplies budgets; statistical experiment design tests information content. The disciplines meet in one workflow: each source-resolved simulation is passed through the measurement operator and recovery analysis, linking physical interpretation to observational tests.

The M3 data-management plan will define provenance and permissions for JRC/national, DWD and CTBTO records, obtained through their original providers under applicable conditions. It will specify budget units/sign conventions, versioned metadata, storage in TB, retention, hosting and maintenance responsibilities through and beyond the fellowship; WP1 verifies transfers and closure by MS1 (M4). Protocols and withheld blocks are versioned before fitting. Curated model products, publication-supporting data and reusable analysis code will be DOI-archived with appropriate licences, FAIR metadata and open formats [19]. Restricted observations remain with authorised users; synthetic examples and tests enable independent reuse.

Sex and gender are not explanatory variables for the atmospheric inventories, fluxes and numerical interventions studied here. The project analyses environmental measurements and model output, with no human exposure, health-outcome or participant data. This assessment will be revisited if the research content changes. Geographic and observational representativeness are addressed through coverage and sampling tests. Inclusive mentoring is described in Section 1.3. The work uses existing records and simulations, with no new radioactive releases or handling.

1.3 Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host

Aarhus University (AU), Department of Environmental Science, Roskilde, will host the project with Zhuyun Ye as main supervisor. Her expertise covers regional Eulerian modelling and data assimilation [13], including co-authorship of the CAMS regional system description [17]. Three previous one-month visits and the joint ^{7}Be–DEHM implementation [7] establish the collaboration. Jesper Heile Christensen, Senior Researcher at AU and a principal DEHM developer [24], will provide complementary mentoring in model development, sensitivity design and model–observation interpretation, drawing on his chemical data-assimilation experience [25].

Period / direction	Specific competence and verifiable output
M1–M4 AU → Chham	Source-tag coding, boundary accounting and numerical verification under Ye's guidance: reviewed code and a passing reconstruction/closure test suite.
M4–M12 AU ↔ Chham	DEHM sensitivities mentored by Christensen; inverse-problem training coordinated by Ye. Output: reviewed recovery/uncertainty protocol before fitting and one methods presentation.
M8–M18 Chham → AU	Radionuclide production, decay, collector conventions and sampling-aware R/Python workflows: two practical sessions with reusable exercises and unit tests.
M18–M24 Chham-led integration	Lead the held-out assessment, developer demonstration and open release; prepare a research portfolio and size-resolved ^{7}Be funding proposal with AU mentoring.

Table 5. Training is assessed through reviewed code, executable methods and independent scientific decisions.

Weekly meetings with Ye and quarterly reviews with Christensen support independent decisions. Chham and Ye will agree a career-development plan in M1 and review it every six months. Chham leads implementation, analysis and documentation; Ye oversees supervision and targeted inverse-problem training. A review before fitting assesses synthetic recovery, uncertainty and scoring. Christensen advises on the follow-on proposal's DEHM component.

Inclusive practice will include transparent authorship/contribution records, equitable access to presenting and leading tasks, invitations that seek gender-diverse participation in methods sessions, accessible hybrid materials and scheduling compatible with caring responsibilities. Feedback on mentoring and workload will be part of the six-month reviews. These measures concern working conditions and knowledge transfer; they do not substitute for the research-content assessment in Section 1.2.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

Essaid Chham is a postdoctoral researcher at Universidad Miguel Hernández, with doctorates in Chemistry (Granada, 2018) and Physics (Abdelmalek Essaâdi, 2019), Erasmus Mundus experience and María Zambrano/APOSTD postdoctoral periods, detailed in Part B2. His combination of Monte Carlo modelling, environmental radioactivity and atmospheric analysis matches the project's integration of source physics, numerical tracers and observations.

His work on sampling intervals [11] informs the observation operator; Open-AMA [12] provides experience in reproducible tools for O2–O3. Research on ^{7}Be prediction [10] and DEHM, Fortran, R/Python and NetCDF skills support implementation. Teaching and supervision are detailed in Part B2. The fellowship adds controlled experiment design and uncertainty-based discrimination of competing explanations.

2. Impact

2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development

Within three to five years, Chham aims to lead an independent research line on atmospheric transport and aerosol processes at a European university or research institute. By M24, he will lead a reusable diagnostic release, target two lead-author manuscript submissions and prepare a follow-on proposal on size-resolved ^{7}Be transport and removal in DEHM. AU mentoring supports its scientific case, partnerships and budget for suitable calls, including the Novo Nordisk Foundation's NERD programme where eligible [26]. Discrimination results guide future observations requiring subsequent funding. Training, developer demonstrations and teaching build leadership; six-month career reviews assess independent decisions, outputs and progress toward this goal.

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities

Dissemination targets atmospheric modellers and environmental-radioactivity researchers; exploitation delivers a usable diagnostic method; communication explains why matching measurements can mask incorrect processes. Demonstrations, training and the data-access interface share the tested benchmark and release, within the existing WP5 allocation.

Audience / timing	Product and uptake measure
Scientific community M15–M24	Two manuscripts on multidecadal source/removal budgets and observational discrimination/transfer. Preprints, open-access routes and at least one conference contribution. Track submissions, DOI releases and feedback.
AU/DEHM developers M12–M24	Demonstrate the benchmark at M12/M22 and deliver a developer note with supported changes and regression tests. Record feedback and reproduction effort; target at least one re-run by a DEHM/CAMS developer outside the project team by M24.
Monitoring/data specialists M18–M24	An observing-value brief on where paired deposition adds information and finer sampling helps. Invite feedback from JRC, DWD and relevant CTBTO/IMS specialists; record responses and reuse requests.
Wider research users / public M8–M24	An open example and data-access interface, two practical training sessions and an accessible radiotracer brief. Record attendance, tutorial completion and use of published outputs.

Table 6. Products, target audiences and measures of use guide dissemination and exploitation throughout the fellowship.

Reusable analysis tools will use an appropriate open-source software licence agreed with AU, respecting ownership and dependency compatibility; host-owned DEHM code remains separately governed. Versioned schemas, tests and a synthetic benchmark support independent reuse. A lightweight interface using established data-serving tools will provide direct downloads of published model subsets selected by period, area, variable and available hourly, daily or monthly resolution, with metadata, version and citation information. Curated products and publication-supporting data remain DOI-archived. Developer demonstrations and an independent re-run will assess usability and guide the final release.

2.3 The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Scientific impact. STRAT-TRACE will link verified ^{7}Be source budgets to observational discrimination of atmospheric explanations. Three wet settings with variable/fixed source forcing over 1986–2024/2025 test robustness across three complete solar cycles. Long European records add temporal reach; CTBTO air observations extend spatial tests within DEHM coverage. Paired concentration–deposition records determine where deposition adds information. The resulting map will identify compatible explanations and the additional sampling needed to distinguish them.

Technological and environmental-assessment pathway. DEHM is one of the 11 models contributing to the CAMS regional air-quality system [17]. Its developers will receive an executable benchmark, a reviewed demonstration and regression tests. Joint concentration–deposition tests will assess whether apparent gains in concentration skill remain consistent with deposition observations, and whether compensating supply–removal errors provide a plausible explanation. The results will guide model development. Usability, re-run effort and independent reproduction will be assessed through developer feedback. Transfer to pollutant forecasting remains a subsequent validation step.

Societal and European relevance. More credible aerosol-process evaluation strengthens the evidence used in air-quality and ecosystem assessments, consistent with the EU Zero Pollution Action Plan [18]. The observing-value brief will help monitoring specialists prioritise where paired concentration–deposition measurements or finer sampling add useful information. FAIR releases and reusable examples [19] make these methods accessible beyond the host team. Wider environmental and economic benefits depend on subsequent adoption and validated transfer to relevant pollutants.

3. Quality and Efficiency of the Implementation

3.1 Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages

The 24-month plan links verified labels, matched experiments over 1986–2024/2025 and independent evaluation of historical and recent observations. AU has confirmed the computational capacity and execution schedule for the full matrix, including fixed-production companions, spin-up and the restart/verification reserve. MS1 at M4 validates source controls, budgets and input continuity before long runs. Reference output follows by M9 and the full response library by M15, leaving time for evaluation and release.

WP / span / effort	Tasks and measurable output	Milestone
WP1 M1–M4 / 3 PM	Verify labels, boundary conversion and spin-up; audit model inputs and JRC/DWD/CTBTO coverage, overlaps and permissions; benchmark resources and lock evaluation blocks. D1: source/measurement protocol and data-management plan, including the historical-input contingency.	MS1, M4: numerical checks pass; inputs and source controls verified; common period and evaluation protocol frozen.
WP2 M3–M9 / 5 PM	Run tagged R and its fixed-production partner over 1986–2024/2025 (78/80 configuration-years if separate); assess solar and regime coverage. D2: paired source-treatment datasets and origin budgets.	MS2, M9: reference budgets verified; totals reconstructed. H1 contrasts follow in WP3.
WP3 M7–M15 / 6 PM	Run W−/W+ and fixed-production partners over the same years (156/160 configuration-years if separate); reweight sources and verify budgets. D3: source–removal response library and first manuscript draft.	MS3, M15: H1/H2 assessed; held-out cases retained.
WP4 M12–M21 / 5 PM	Recovery and locked historical/recent tests; assess CTBTO spatial transfer with air records and joint skill with valid air–deposition pairs. Check stable segments and source controls. D4: discrimination map, observing brief and second manuscript draft.	MS4, M21: O3 classified, including explicit non-discrimination bounds.
WP5 M1–M24 / 5 PM	Training, supervision, data stewardship, developer engagement and dissemination. D5: licensed tools, data-access interface, methods materials, career portfolio and follow-on funding proposal.	MS5, M24: releases complete; two manuscript submissions targeted.

Table 7. Total effort is 24 person-months (PM), distributed across overlapping work packages. WP2 begins with preparation; long production runs follow MS1. Deliverables and milestones retain a single-researcher workload.

WP	1–3	4–6	7–9	10–12	13–15	16–18	19–21	22–24
WP1	●	●						
WP2	●	●	●					
WP3			●	●	●			
WP4				●	●	●	●	
WP5	●	●	●	●	●	●	●	●
Gate / output		MS1	MS2		MS3		MS4	MS5

Gantt chart. Filled cells indicate activity during all or part of each interval. D1–D5 correspond to MS1–MS5; the data-management plan is due in M3.

Execution budget and control of scope

Three wet settings over 39/40 years require 117/120 configuration-years with variable production. Fixed partners run concurrently as additional species where supported; separate execution doubles the total to 234/240. AU's confirmed provision includes pre-1986 spin-up and a separate 20% restart/verification reserve. WP1 benchmarks contrasting seasonal and wet/dry months with all 14 tracers, online budgets and input/output to refine scheduling and storage. All settings share one meteorological archive.

Delivery risks and scientific decision branches

The risk plan protects delivery while preserving the interpretation of inconclusive tests. A regime with weak discrimination still yields measured response limits and an observing-value assessment; inadequate information across the whole network would constrain the empirical contribution. The qualitative ratings below will be updated at each milestone.

Risk / residual likelihood / impact	Trigger, mitigation and retained result
Numerical closure or linearity fails Medium / High	Failed ledger or reconstruction tests stop source-fraction claims. Audit boundary, limiter and decay terms; use a verified coarser partition or source-restricted reruns within the costed reserve. Report the numerical limitation if closure cannot be established.
Paired records or metadata inadequate Medium / High	Audit JRC/DWD/CTBTO coverage, overlap, permissions and method eras. Retain valid pairs for joint tests and stable segments for temporal comparisons. CTBTO adds air tests; delivery delays do not displace the established paired baseline. Redesign if core paired coverage is inadequate.
Little discrimination across the available network Medium / Medium	Early synthetic tests evaluate the actual observing design. Report compatible scenarios and tested discrimination bounds; quantify the information added by alternative sampling. Preserve verified source-budget results and the geographic scope of inference.
Historical-input gaps or workflow delays Medium / High	If historical inputs cannot be completed by MS1, all wet settings and both source treatments use the longest verified common continuous period, prioritising the 2009–2024 observational baseline. Update solar-cycle coverage and evaluation blocks together. Checkpoints, online sums and the allocated reserve address workflow interruptions.
Candidate improvement fails held-out tests Medium / Medium	Separate air/deposition scores and source/discrepancy tests expose compensation. Deliver the rejected explanation and its regression test; retain the locked validation blocks.

Table 8. Qualitative residual assessments will be reviewed at each milestone. The response library and discrimination bounds remain deliverables when a particular explanation is rejected.

3.2 Quality and capacity of the host institutions and participating organisations, including hosting arrangements

AU's Department of Environmental Science develops atmospheric transport, chemistry and deposition models across scales [14,15]. Its DEHM contribution to CAMS [17] and the joint ^{7}Be implementation [7] give the host the scientific expertise and developer connections needed for this project. The reduced-chemistry research configuration supports controlled diagnostics, with any later operational changes assessed in their relevant pollutant and forecasting configurations.

AU's confirmed provision supports the complete matrix and M9/M15 schedule. Chham will assemble the 1986–2024/2025 inputs, archive the CRAC-derived upper-boundary formulation and fields, and benchmark the allocated resources in WP1. Ye will oversee execution and scientific decisions. The signed CTBTO agreement complements JRC and DWD data access; WP1 verifies the usable records and permissions. Long runs follow MS1 source-control and input-continuity checks; historical contingencies retain a common period across configurations and source treatments.

Hourly receptor series and online pressure-band/regional budgets cover both source treatments over the full period; detailed 3-D fields are retained for selected episodes. At approximately 6 MB per hourly file, continuous output over 234/240 configuration-years would occupy 12–13 TB for that variable set. AU infrastructure provides over 600 TB of storage capacity. WP1 scales the estimate to retained tagged fields and episodes, budgeting meteorological inputs, checkpoints and copies separately. The M3 plan records TB requirements and retention, updated with pilot results. Curated daily/monthly products reduce download volumes, using time-weighted concentration means and integrated deposition; published subsets are directly accessible through the WP5 interface and repository.

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